CSS Quantum Codes from Surface Tilings

A Lean 4 Formalisation

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Abstract

A tiling of a closed orientable surface of genus g by polygons gives a CSS (Calderbank–Shor–Steane) quantum error-correcting code whose physical qubits correspond to edges, X-stabilizers to vertices, and Z-stabilizers to faces. The number of logical qubits is $k = 2g$, the first Betti number of the surface. We give a self-contained proof and describe a complete formalisation in Lean 4 (sorry-free for the core pipeline), with concrete instances including the Heawood graph on the torus ($g = 1$, $k = 2$), the Möbius–Kantor graph on the Bolza surface ($g = 2$, $k = 4$), the Klein quartic ($g = 3$, $k = 6$), and the Clayworth graph from the exceptional Lie group $G_2(2)$ on a surface of genus 505 ($k = 1010$).

1. Introduction

A CSS quantum code [1, 2] is specified by two binary parity-check matrices H_X and H_Z satisfying $H_X H_Z^\top = 0$. Rows of H_X generate the X-stabilizers; rows of H_Z generate the Z-stabilizers. The code encodes

$$k = n - \operatorname{rank}(H_X) - \operatorname{rank}(H_Z)$$

logical qubits into n physical qubits.

Homological quantum codes [3] construct H_X and H_Z from a chain complex. The simplest instance: tile a closed orientable surface of genus g by polygons, form the cellular chain complex

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

over $\mathbb{F}_2$, set $H_X = \partial_1$ (vertex–edge incidence) and $H_Z = \partial_2^\top$ (face–edge incidence). The chain complex condition $\partial_1 \circ \partial_2 = 0$ gives the CSS orthogonality $H_X H_Z^\top = 0$, and Euler's formula forces $k = 2g$.

This note gives a self-contained account of this construction and its formalisation in Lean 4. The connection to the broader programme linking spectral theory, sheaf cohomology, and LLM interpretability [5, 6] is discussed in Section 6.

2. The construction

2.1. Surface tilings and boundary operators

A *surface tiling* is a 2-cell embedding of a graph on a closed orientable surface: vertices (0-cells), edges (1-cells), and faces (2-cells), where each face boundary is a closed directed trail in the graph. Let V, E, F denote the numbers of vertices, edges, and faces.

The *boundary operators* over $\mathbb{F}_2$ are:

- $\partial_1 \in \mathbb{F}_2^{V \times E}$: the vertex–edge incidence matrix. Entry $\partial_1(v, e) = 1$ if v is an endpoint of e, else 0. (Over $\mathbb{F}_2$, loops would give $1 + 1 = 0$; we exclude loops.)
- $\partial_2 \in \mathbb{F}_2^{E \times F}$: the edge–face boundary matrix. Entry $\partial_2(e, f) = 1$ if edge e appears in the boundary walk of face f, else 0. (Faces are trails, so each edge appears at most once per face.)

2.2. The chain complex condition: $\partial_1 \circ \partial_2 = 0$

Proposition 1. *For any surface tiling, $\partial_1 \cdot \partial_2 = 0$ over $\mathbb{F}_2$.*

Proof. Fix a vertex v and a face f with boundary walk $e_0, e_1, \ldots, e_{r-1}$. Each step i of the walk starts at some vertex and ends at another. The closed walk condition says that $\text{end}(i) = \text{start}(i{+}1 \bmod r)$.

The product $(\partial_1 \cdot \partial_2)(v, f) = \sum_i \partial_1(v, e_i)$. Now $\partial_1(v, e_i) = [\text{start}(i) = v] + [\text{end}(i) = v]$ regardless of traversal direction (addition is commutative). Split the sum:

$$\sum_i [\text{start}(i) = v] + \sum_i [\text{end}(i) = v].$$

By the closed walk condition plus the fact that cyclic rotation $i \mapsto i{+}1 \bmod r$ is a bijection on $\{0, \ldots, r{-}1\}$:

$$\sum_i [\text{end}(i) = v] = \sum_i [\text{start}(i{+}1 \bmod r) = v] = \sum_i [\text{start}(i) = v].$$

Therefore $(\partial_1 \cdot \partial_2)(v, f) = 2 \sum_i [\text{start}(i) = v] = 0$ in $\mathbb{F}_2$. □

3. The main theorem: $k = 2g$

Theorem 2. *Let T be a surface tiling of a closed orientable surface of genus g with V vertices, E edges, and F faces. Set $H_X = \partial_1$ and $H_Z = \partial_2^\top$. If the primal graph is connected and the dual graph is connected (equivalently, each edge borders exactly two distinct faces), then the resulting CSS code encodes exactly $k = 2g$ logical qubits.*

The proof has two ingredients: rank computations and Euler's formula.

3.1. Rank of ∂_1: the kernel characterisation

Lemma 3. *If the primal graph is connected and $V \geq 2$, then* $\text{rank}(\partial_1) = V - 1$.

Proof. Upper bound. Each column of ∂_1 has exactly two nonzero entries (the two endpoints), so each column sums to $1 + 1 = 0$ in $\mathbb{F}_2$. Therefore the all-ones vector $\mathbf{1}_V$ lies in $\ker(\partial_1^\top)$, giving $\dim \ker(\partial_1^\top) \geq 1$, hence $\text{rank}(\partial_1) \leq V - 1$ by rank–nullity.

Lower bound. We show $\ker(\partial_1^\top) = \text{span}\{\mathbf{1}_V\}$. Let $x \in \ker(\partial_1^\top)$. Then for every edge e with endpoints u, v:

$$(\partial_1^\top x)(e) = x(u) + x(v) = 0 \implies x(u) = x(v).$$

By connectivity, for any two vertices u, w there is a walk $u = v_0, v_1, \ldots, v_\ell = w$, and the equation propagates: $x(u) = x(v_1) = \cdots = x(w)$. So x is constant on all vertices. Over $\mathbb{F}_2$, constant means $x = 0$ or $x = \mathbf{1}_V$, so $\ker(\partial_1^\top) = \text{span}\{\mathbf{1}_V\}$ has dimension 1. Rank–nullity gives $\text{rank}(\partial_1) = V - 1$. □

3.2. Rank of ∂_2: the dual kernel characterisation

Lemma 4. *If each edge borders exactly two faces and the dual graph is connected with $F \geq 2$, then* $\operatorname{rank}(\partial_2) = F - 1$.

Proof. The argument mirrors Lemma 3. The two-sides hypothesis implies $\partial_2 \cdot \mathbf{1}_F = 0$ (each row of ∂_2 has exactly two nonzero entries), giving the upper bound. For the lower bound: if $y \in \ker(\partial_2)$, then for each edge e bordering faces f_1, f_2, we get $y(f_1) + y(f_2) = 0$, i.e., $y(f_1) = y(f_2)$. Dual connectivity propagates this to all faces, so $\ker(\partial_2) = \operatorname{span}\{\mathbf{1}_F\}$. □

3.3. Euler arithmetic

Proof of Theorem 2. By Lemmas 3 and 4:

$$\begin{aligned} k &= E - \operatorname{rank}(H_X) - \operatorname{rank}(H_Z) \\ &= E - (V - 1) - (F - 1) \\ &= E - V - F + 2 \\ &= 2g, \end{aligned}$$

where the last step uses Euler's formula $V - E + F = 2 - 2g$ for a closed orientable surface of genus g. □

4. Concrete instances

Every surface tiling from a triangle group $\Delta(2, 3, r)$ acting on a finite group G gives an instance. The construction (formalised in `TriangleGroupSurface.lean`) uses the Sabidussi coset representation [4]: vertices are cosets of $\langle S \rangle$ (order 3), edges are cosets of $\langle T \rangle$ (order 2, where $T = RS$), and faces are cosets of $\langle R \rangle$ (order r). The genus follows from Euler's formula: $g = 1 - |G|/2 \cdot (1/2 + 1/3 + 1/r - 1)$.

Surface	Graph	G	Δ	g	CSS	Lean
Torus	Heawood	G_{42}	$(2, 3, 6)$	1	$[\![21, 2]\!]$	sorry-free
Bolza	Möb.-Kantor	$\mathrm{GL}(2, 3)$	$(2, 3, 8)$	2	$[\![24, 4]\!]$	sorry-free
Klein quartic	F056A	$\mathrm{PSL}(2, 7)$	$(2, 3, 7)$	3	$[\![84, 6]\!]$	3 data sorry
Clayworth	4032v	$G_2(2)$	$(2, 3, 12)$	505	$[\![6048, 1010]\!]$	rank hyps

Table 1: CSS codes from triangle group tilings. "Sorry-free" means every axiom is verified by the Lean 4 kernel. The Klein quartic has 3 data sorrys (verified in GAP). The Clayworth code has rank hypotheses verified by Gaussian elimination over $\mathbb{F}_2$ in Python.

The Heawood graph is the Levi graph (incidence graph) of the Fano plane $\mathrm{PG}(2, 2)$, embedded on the torus with 7 hexagonal faces in the $\{6, 3\}$ tiling.

The Möbius–Kantor graph $\mathrm{GP}(8, 3)$ (the generalised Petersen graph) embeds on the Bolza surface—the most symmetric genus-2 surface—with 6 octagonal faces in the $\{8, 3\}$ tiling.

The Klein quartic has the maximum number of automorphisms for genus 3 (Hurwitz bound: $|\mathrm{Aut}| = 168 = 84(g - 1)$). Its $\{7, 3\}$ tiling gives 24 heptagonal faces.

The Clayworth graph, constructed from the exceptional Lie group $G_2(2)$ (order 12096), gives the largest code in this family: 6048 physical qubits encoding 1010 logical qubits on a surface of genus 505.

Remark 5. *Not every Sabidussi graph admits a tiling. The Zhou graph* $\mathrm{Sab}(\mathrm{PSL}(2,13), S_3)$ *(Foster census F182A, 182 vertices) does not arise from a triangle group of the form* $\Delta(2,3,r)$ *and does not tile a surface in this way.*

5. The Lean formalisation

The formalisation consists of the following files, building a sorry-free proof chain for the core theorem:

1. `Basic.lean` — defines `BinaryLinearCode` (a parity-check matrix over $\mathbb{F}_2$ and its kernel as code space) and proves $\dim = n - \mathrm{rank}(H)$ via rank–nullity.

2. `CSSCode.lean` — defines `SurfaceTiling` (V, E, F, g with Euler's formula as an axiom) and `CSSFromTiling` (H_X, H_Z with the orthogonality condition $H_X H_Z^\top = 0$). States and proves `css_k_eq_2g`: given rank hypotheses, $k = 2g$ follows by `linarith` from Euler's formula.

3. `CellularSurface.lean` — defines the combinatorial data of a 2-cell embedding: edge endpoints, face boundary walks (as injective sequences of directed edges satisfying a closed walk axiom), boundary operators ∂_1 and ∂_2 as matrices over $\mathbb{F}_2$. **Proves** $\partial_1 \cdot \partial_2 = 0$ (sorry-free). The proof follows Proposition 1: reindex via the cyclic bijection, then use $x + x = 0$ in $\mathbb{F}_2$.

4. `IncMatrixRank.lean` — proves the rank theorems. Upper bounds via the all-ones kernel vector. Lower bounds via the kernel characterisation: walks in connected graphs propagate equality of kernel elements, forcing $\ker(\partial_1^\top) = \mathrm{span}\{\mathbf{1}_V\}$. The dual argument uses dual adjacency (two faces sharing an edge) and the two-sides hypothesis. **All sorry-free.**

5. `SurfaceCSS.lean` — assembly: constructs `CSSFromTiling` from a `CellularSurface` ($H_X = \partial_1$, $H_Z = \partial_2^\top$, orthogonality from $\partial_1 \cdot \partial_2 = 0$) and proves `css_k_eq_2g_from_surface`. Hypotheses: Euler, primal connected, dual connected, two-sides. **Sorry-free.**

The triangle group construction (`TriangleGroupSurface.lean`) provides a systematic way to generate concrete instances from finite groups. Darts (group elements under the regular representation) are partitioned into vertices, edges, and faces via cosets of $\langle S\rangle$, $\langle T\rangle$, $\langle R\rangle$ respectively. The closed walk axiom follows from $T = RS$ and S preserving vertex cosets: $\mathrm{vertex}(T(d)) = \mathrm{vertex}(S(R(d))) = \mathrm{vertex}(R(d))$.

6. Connection to spectral theory and sheaf cohomology

The number $2g$ is one object wearing three hats:

1. **Quantum codes.** The number of logical qubits: $k = E - \mathrm{rank}(H_X) - \mathrm{rank}(H_Z) = 2g$, proved in Theorem 2.

2. **Hodge theory.** The space of harmonic 1-forms has dimension $\dim \ker \Delta_1 = 2g$, where Δ_1 is the Laplace–Beltrami operator on 1-forms. By the Hodge theorem, $\ker \Delta_1 \cong H^1(M;\mathbb{R})$.

3. **Sheaf cohomology.** In a presheaf model of local linearisations (e.g., SAE decompositions of neural network activations), $\dim H^1$ counts the independent obstructions to gluing local descriptions into a global one [5]. The failure to glue is polysemanticity.

All three are manifestations of the first Betti number $\beta_1 = \dim H_1(M;\mathbb{R}) = 2g$, which counts the "handles" of the surface. The chain complex $\partial^2 = 0$ that gives CSS orthogonality is the same chain complex whose homology defines H_1; the harmonic forms are its Hodge-theoretic shadow; the sheaf H^1 is its re-interpretation in the language of local-to-global obstructions.

This triangle—codes, spectra, sheaves—is developed further in [5, 6].

References

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